

Final Exam

Exercise 1. Find the change of coordinates matrix from the basis $1, 1 + t$ to the basis $1 - t, 2t$ in $\mathbb{P}_1$.

Exercise 2. Let $Perm(n)$ be the set of permutations of $\{1, 2, \dots, n\}$. Let $\tau \in Perm(n)$. Show that the map from $Perm(n)$ to $Perm(n)$ defined by $\sigma \mapsto \sigma\tau$ is a bijection (one-to-one and onto). In other words, composition with a fixed permutation τ takes the list of all permutations and rearranges it, but does not duplicate or remove any (it permutes the permutations).

Exercise 3. Show that the determinant of a matrix is the product of its eigenvalues (counting multiplicities). **Hint:** First show that the characteristic polynomial $\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda)$ (note the λ_i 's occur before λ), where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (with multiplicity). Then plug in $\lambda = 0$. Note that in general, a polynomial of degree n can be written as $p(\lambda) = c(\lambda - r_1)(\lambda - r_2) \cdots (\lambda - r_n)$ (note the λ comes before each λ_i), so the first step is to show that this c is $(-1)^n$. If you use the permutation definition of the determinant, there is exactly one term that is of degree n in λ , and it comes from the identity permutation.

Exercise 4. Suppose the entries of A and A^{-1} are integers. What are the possible values of $\det(A)$? **Hint:** Use the fact that $\det(A) \cdot \det(A^{-1}) = 1$.